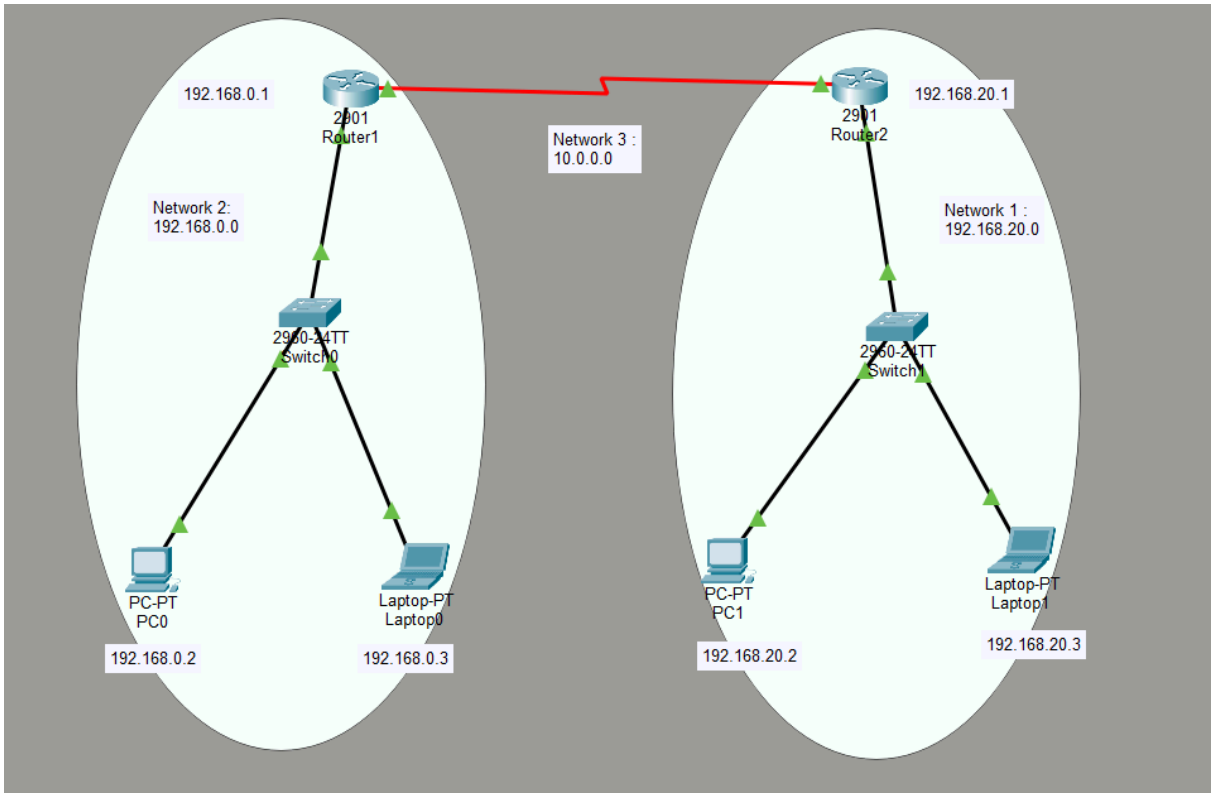
1) Router Configuration and RIP



Circuit 1 : For Router Ip Settings and Configure Routing information Protocol (RIP)

Network Name	Network ID	Devices Connected	Cable Communication
Network 1	192.168.0.0	192.168.0.2 192.168.0.3	fastEthernet Coaxial cable
Network 2	10.0.0.0	10.0.0.2 10.0.0.1	Serial DCE
Network 3	192.168.20.0	192.168.20.2 192.168.20.3	fastEthernet Coaxial cable

Router Configuration :
Router R1 :

```
Router>en
Router#conf
Router#conf terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#int
R1(config)#interface gig
R1(config)#interface gig
R1(config)#interface gigabitEthernet 0/0
R1(config)#interface gigabitEthernet 0/0
R1(config-if)#ip addr 192.168.0.1 255.255.255.0
R1(config-if)#no shut

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#exit
R1(config)#int ser
R1(config)#int serial 0/3/0
R1(config-if)#ip addr 10.0.0.1 255.0.0.0
R1(config-if)#no shut

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
R1(config-if)#
R1(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

R1(config-if)#exit
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#network 10.0.0.0
R1(config-router)#network 192.168.0.0
R1(config-router)#exit
R1(config)#
```

Two Interfaces of the Router R1 :

giggabitEthernet 0/0 : 192.168.0.1
Serial 0/3/0 : 10.0.0.1

Connected Device to Router R1 at interface giggabitEthernet 0/0 : (Network 1)

PC0 : 192.168.0.2
Laptop0 : 192.168.0.3

Router R2 :

Two Interfaces of the Router R2 :

giggabitEthernet 0/0 : 192.168.20.1
Serial 0/3/0 : 10.0.0.2

Connected Devices to Router R2 at interface giggabitEthernet 0/0 : (Network 3)

PC1 : 192.168.20.2
Laptop1 : 192.168.20.3

```
Router>en
Router#conf t
Router#conf terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#int gig
R2(config)#int gigabitEthernet 0/0
R2(config-if)#ip addr 192.168.20.1 255.255.255.0
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R2(config-if)#exit
R2(config)#int ser
R2(config)#int serial 0/3/0
R2(config-if)#ip addr 10.0.0.2 255.0.0.0
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

R2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
exit
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#network 10.0.0.0
R2(config-router)#network 192.168.20.0
R2(config-router)#no ip
R2(config-router)#no ip sum
R2(config-router)#no ip sum
R2(config-router)#no ip-sum
R2(config-router)#no ip-sum
R2(config-router)#
R2(config-router)#exit
R2(config)#
R2(config)#
```

Pinging From PC0 to Network 3 Devices …

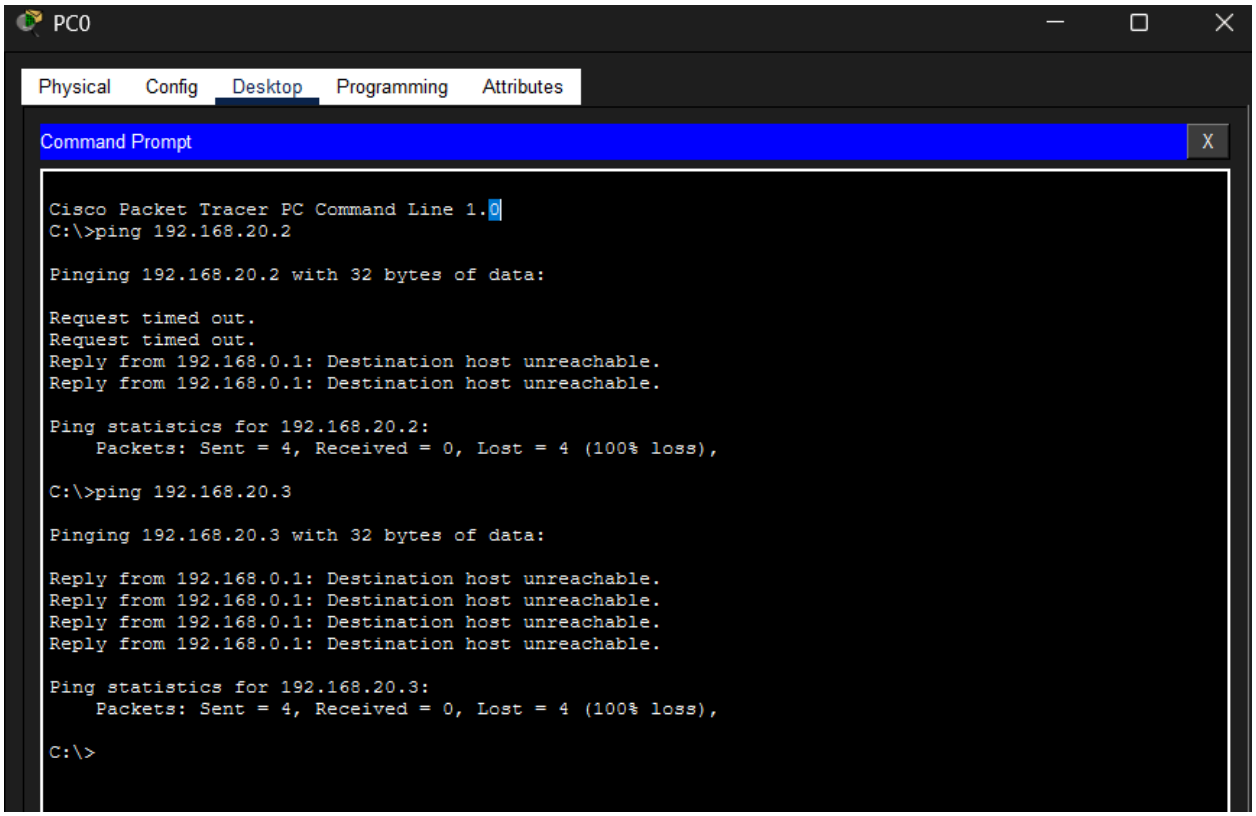
```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time=15ms TTL=254
Reply from 192.168.20.1: bytes=32 time=10ms TTL=254

Ping statistics for 192.168.20.1:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 15ms, Average = 12ms

Control-C
^C
C:\>ping 192.168.20.3

Pinging 192.168.20.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.3: bytes=32 time=11ms TTL=126
Reply from 192.168.20.3: bytes=32 time=12ms TTL=126
Reply from 192.168.20.3: bytes=32 time=22ms TTL=126

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 22ms, Average = 15ms

C:\>
```

2) Access Control List

1. Standard Access Control List :

Created a Access list with number 10 , deny all traffic going out from 192.168.0.0 to 192.168.0.255

```
R1#show access-lists
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#access-list 10 deny 192.168.0.0 0.0.0.288
R1(config)#access-list 10 deny 192.168.0.0 0.0.0.2255
R1(config)#access-list 10 deny 192.168.0.0 0.0.0.255
R1(config)#access-list 10 permit any
R1(config)#int ser
R1(config)#int serial 0/3/0
R1(config-if)#ip access-group 10 out
R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#write memory
Building configuration...
[OK]
R1#
```

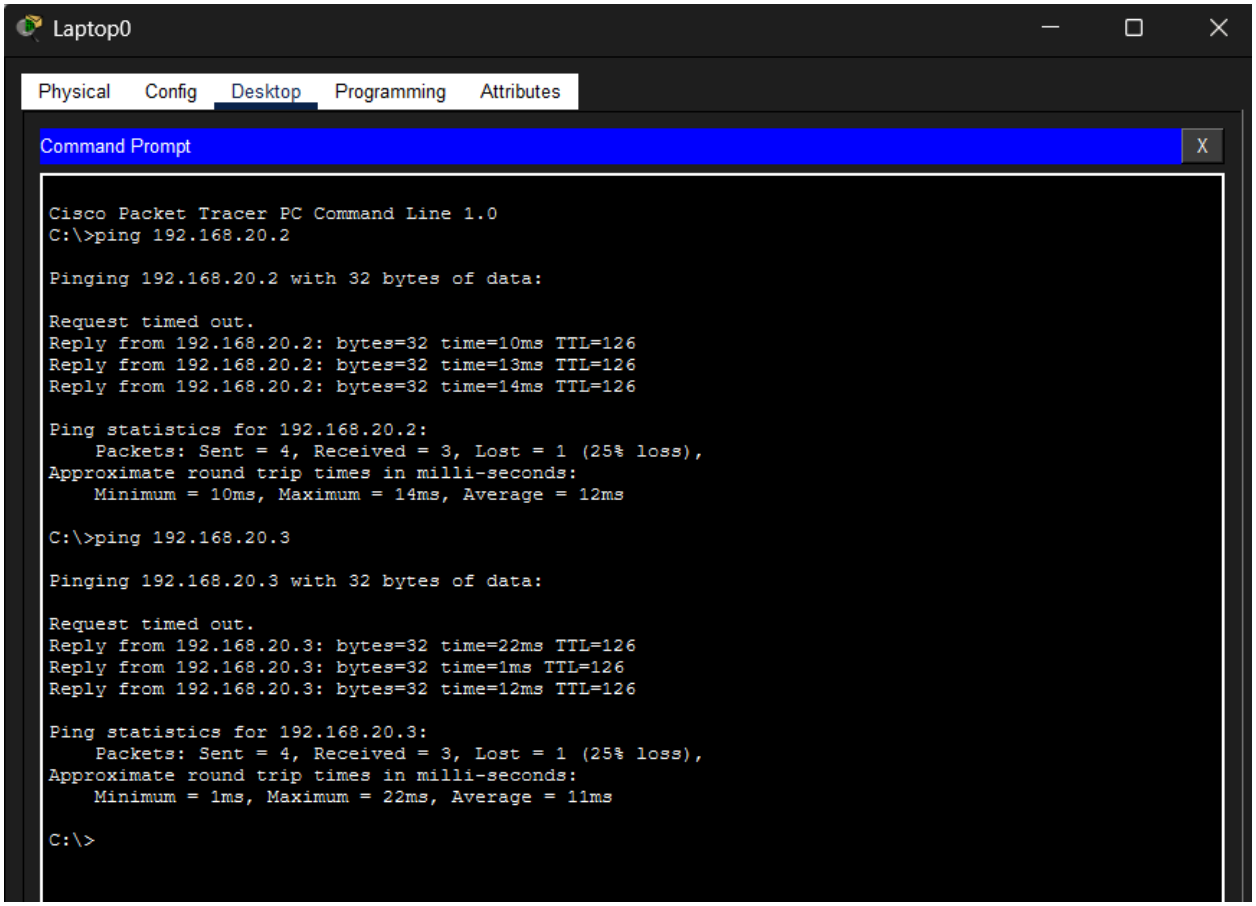


2. Extended Access Control List :

Deny all the traffic from one ip to another with specifications as port and protocols.

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#access-list 100 deny icmp host 192.168.0.2 host 192.168.20.2
R1(config)#access-list 100 deny tcp host 192.168.0.2 host 192.168.20.3 eq 80
R1(config)#access-list 100 permit ip any any
R1(config)#int ser
R1(config)#int serial
% Incomplete command.
R1(config)#int serial 0/3/0
R1(config-if)#ip access-group 100 in
R1(config-if)#exit
R1(config)#en
% Ambiguous command: "en"
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#w
Building configuration...
[OK]
R1#show access-lists
Standard IP access list 10
    10 deny 192.168.0.0 0.0.0.255
    20 permit any
Extended IP access list 100
    10 deny icmp host 192.168.0.2 host 192.168.20.2
    20 deny tcp host 192.168.0.2 host 192.168.20.3 eq www
    30 permit ip any any (1 match(es))

R1#
R1#
```

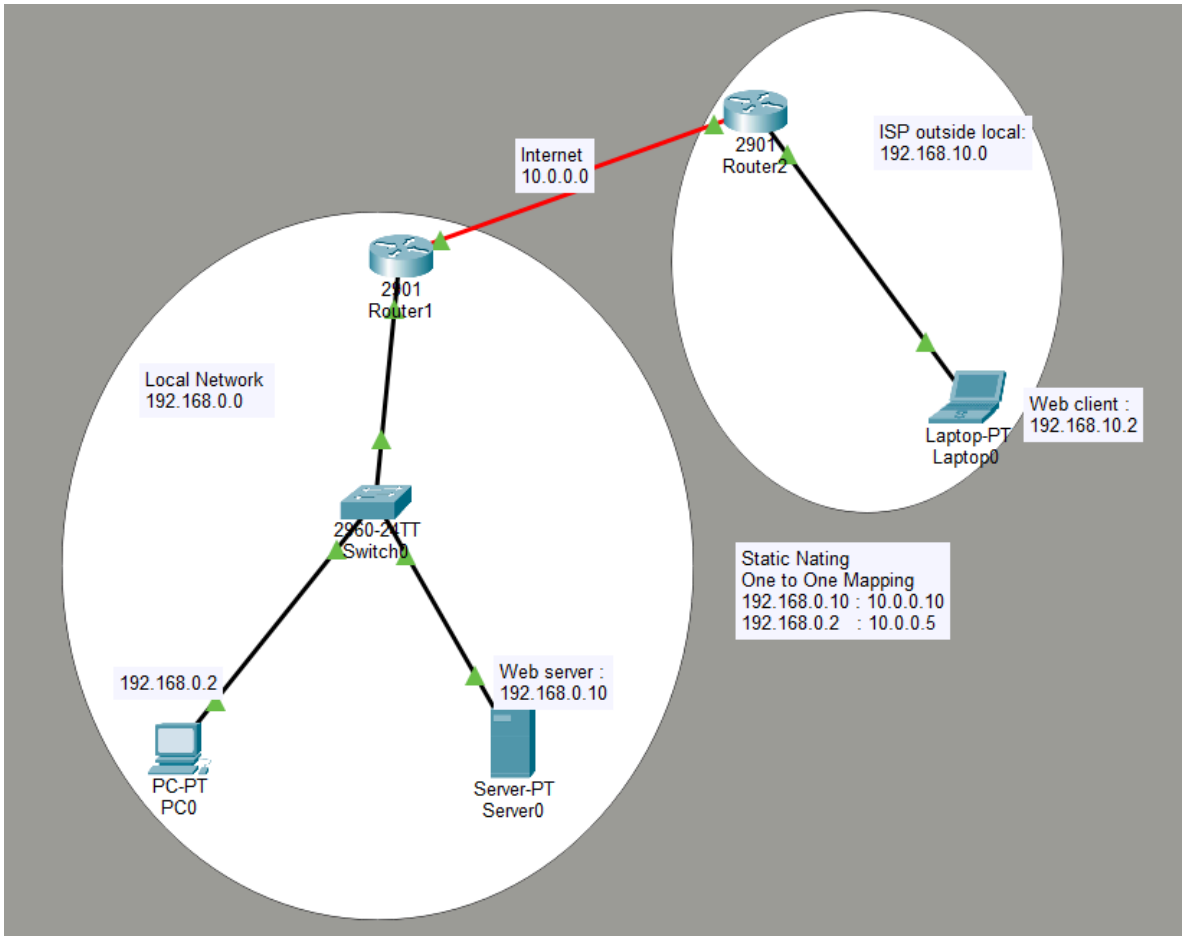
```
R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#no access-list 10
R1(config)#access-list 10 deny host 192.168.0.2
R1(config)#access-list 10 permit any
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
w
Building configuration...
[OK]
R1#show access-list
Extended IP access list 100
    10 deny icmp host 192.168.0.2 host 192.168.20.2
    20 deny tcp host 192.168.0.2 host 192.168.20.3 eq www
    30 permit ip any any (6 match(es))
Standard IP access list 10
    10 deny host 192.168.0.2
    20 permit any

R1#
```



3) Network Address Translation (NAT) :

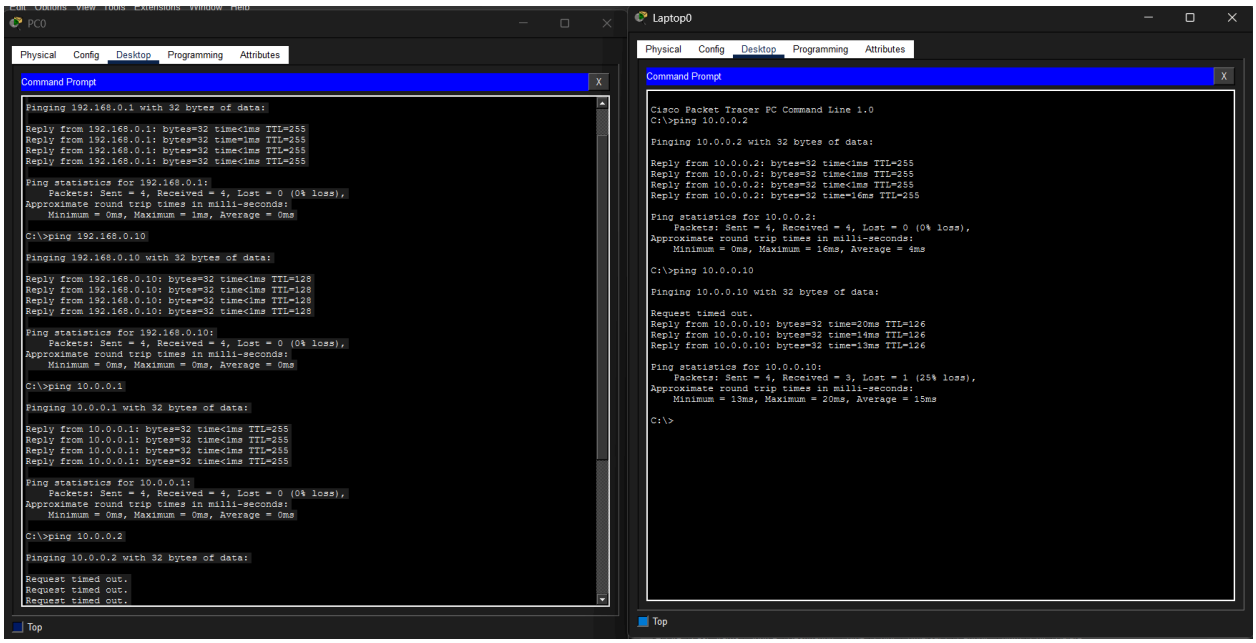
Network Name	Network ID	Devices Connected	Cable Communication
Inside Local Network	192.168.0.0	192.168.0.2 192.168.0.10	fastEthernet Coaxial cable
Internet	10.0.0.0	10.0.0.2 10.0.0.1	Serial DCE
Outside Local Network	192.168.10.0	192.168.10.2	fastEthernet Coaxial cable

1. Static NAT : One to One mapping of IPs, increase as the Number of devices in the network increase.




```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip nat inside source static 192.168.0.10 10.0.0.10
Router(config)#int gig
Router(config)#int gigabitEthernet
% Incomplete command.
Router(config)#int gigabitEthernet 0/0
Router(config-if)#ip addr 192.168.0.1 255.255.255.0
Router(config-if)#ip nat inside
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int seri
Router(config)#int serial
% Incomplete command.
Router(config)#int serial  0/3/0
Router(config-if)#ip addr 10.0.0.1 255.255.255.252
Router(config-if)#ip nat outside
Router(config-if)#no shut
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#exit
Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.2
Router(config)#
```

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int ser
Router(config)#int serial 0/3/0
Router(config-if)#ip addr 10.0.0.2 255.255.255.252
Router(config-if)#no sut
                      ^
% Invalid input detected at '^' marker.

Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

Router(config-if)#exit
Router(config)#ip route 10.0.0.
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, chan
Router(config)#ip route 10.0.0.10 255.255.255.255 10.0.0.1
Router(config)#
```



2. Dynamic NAT

Many to One mapping , reduces the use of ips , and provides a range of ip pool to select the ips from.

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no ip nat inside source static 192.168.0.2 10.0.0.3
Router(config)#ip nat inside source list 1 pool Mypool
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router#show ip nat translation
Pro  Inside global     Inside local       Outside local      Outside global
icmp 10.0.0.5:5        192.168.0.2:5      10.0.0.2:5         10.0.0.2:5
icmp 10.0.0.5:6        192.168.0.2:6      10.0.0.2:6         10.0.0.2:6
icmp 10.0.0.5:7        192.168.0.2:7      10.0.0.2:7         10.0.0.2:7
icmp 10.0.0.5:8        192.168.0.2:8      10.0.0.2:8         10.0.0.2:8
icmp 10.0.0.6:5        192.168.0.10:5     10.0.0.2:5         10.0.0.2:5
icmp 10.0.0.6:6        192.168.0.10:6     10.0.0.2:6         10.0.0.2:6
icmp 10.0.0.6:7        192.168.0.10:7     10.0.0.2:7         10.0.0.2:7
icmp 10.0.0.6:8        192.168.0.10:8     10.0.0.2:8         10.0.0.2:8

Router#
```

3. PAT : Port Address Translation

Instead of using a number of ips , we can use different port numbers . One ip but two different port numbers separates the communication between devices.

```
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up


Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no ip nat inside source list 1 pool Mypool
Router(config)#ip nat inside source list
Router(config)#ip nat inside source list 1 interface sr
Router(config)#ip nat inside source list 1 interface ser
Router(config)#ip nat inside source list 1 interface serial 0/3/0 overload
Router(config)#clear ip nat translation
                ^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#clear ip nat translation *
Router#show ip nat translation
Pro  Inside global     Inside local       Outside local      Outside global
icmp 10.0.0.1:1024     192.168.0.10:1     10.0.0.2:1         10.0.0.2:1024
icmp 10.0.0.1:1025     192.168.0.10:2     10.0.0.2:2         10.0.0.2:1025
icmp 10.0.0.1:1026     192.168.0.10:3     10.0.0.2:3         10.0.0.2:1026
icmp 10.0.0.1:1027     192.168.0.10:4     10.0.0.2:4         10.0.0.2:1027

Router#show ip nat translation
Pro  Inside global     Inside local       Outside local      Outside global
icmp 10.0.0.1:5        192.168.0.2:5      10.0.0.2:5         10.0.0.2:5
icmp 10.0.0.1:6        192.168.0.2:6      10.0.0.2:6         10.0.0.2:6
icmp 10.0.0.1:7        192.168.0.2:7      10.0.0.2:7         10.0.0.2:7
icmp 10.0.0.1:8        192.168.0.2:8      10.0.0.2:8         10.0.0.2:8

Router#
```